Harnessing LLMs for Medical Insights: NER Extraction from Summarized Medical Text

Problem Statement: The paper talks about to produce concise, coherent summaries of medical literature, prioritizing clinical relevance. Automated NER is adopted to extract vital entities from the summaries generated by LLMs.

Algorithms Used: More research is undertaken toward NER techniques. PEGASUS LLM from Google – MLM(Encoder-Masked Language Model) and GSG(Decoder-Gap Sentence Generation), T5 (transformer based encoder-decoder model for supervised and self-supervised learning) and GPT-3.

Datasets: PubMed Database managed by NCBI. Accessible via E-utilities, that supports XML and JSON format for searching and retrieving records.

Model Training and Testing: Models are trained for 10 epochs. *ROUGE i.e. Recall-Oriented Understudy for Gisting Evaluation* and BERT metrics has been used to evaluate the summaries produced by the models.

Results: Models performance comparison for metrics evaluation is as follows

Model Name	ROUGE-1	ROUGE-2	ROUGE-L	BERTScore_Precision	BERTScore_Recall	BERTScore_F1score
T5	0.4008	0.1350	0.2426	0.6314	0.6240	0.6256
Pegasus	0.4402	0.1742	0.2506	0.6378	0.6610	0.6468
GPT3.5	0.4573	0.1783	0.2784	0.6565	0.6617	0.6585

Conclusions: This study presented large language models and named entity recognition to summarize and extract information from the vast PubMed medical literature database. State-of-the-art LLMs like T5 were fine-tuned to generate concise summaries, while a BERT-based NER model identified critical biomedical entities within the summaries.

Relevance to Our Team: NER techniques used could be applied in the model. Performance of the model could be improvised in pre-processing.

Reference:

https://ieeexplore.ieee.org/document/10724860

<u>Biomedical Chat Assistant with Personalized Document Reader Using BioMistral and RAG</u>

Problem Statement: This paper reports the development of a Biomedical Chat Assistant equipped with a personalized reader for documents, using the capabilities provided by the BioMistral framework and Retrieval-Augmented Generation techniques.

Algorithms Used: BioMistral LLM integrated with RAG. BioMistral supports in understanding biomedical text. Document chunking pubmedbert.

Datasets: PDF documents on clinical research

Results: Biomedical Chat Assistant by analyzing the various established methods such as PubMed Search API, BioBERT, Clinical BERT, and Semantic Scholar.

Metric	Biomedical Chat Assistant	PubMed Search API	BioBERT	Clinical BERT	Semantic Scholar
Accuracy	92%	75%	80%	78%	76%
Response Time	1.5 seconds	4 seconds	3 seconds	3.5 seconds	5 seconds
Precision	88%	70%	75%	72%	73%
Recall	85%	60%	70%	65%	68%
F1 Score	0.86	0.67	0.77	0.72	0.71
User Satisfaction Rating	4.5/5	3.5/5	3.8/5	3.6/5	3.7/5
User Engagement	7 interactions/session	3-4 interactions/session	3-4 interactions/session	3-4 interactions/session	3-4 interactions/session

Conclusions: The system efficiently processed numerous unstructured PDF medical reports to retrieve relevant information with high accuracy.

Open Challenges: Further refinements are needed to address challenges like document format variability and medical language complexity.

Relevance to Our Team: BioMistral LLM and RAG, Chroma vector to store the Vector embeddings and Recursive Text Splitter for document chunking can be relevant to our project.

Reference:

https://ieeexplore.ieee.org/document/11020219

The Impact of LoRA Adapters on LLMs for Clinical Text Classification Under Computational and Data Constraints

Problem Statement: Fine-tuning Large Language Models (LLMs) for clinical Natural Language Processing (NLP) poses significant challenges due to domain gap, limited data, and stringent hardware constraints. In this paper, the four adapter techniques—Adapter, Lightweight, TinyAttention, and Gated Residual Network (GRN)- equivalent to Low-Rank Adaptation (LoRA), for clinical note classification under real world, resource-constrained conditions are evaluated.

Algorithms Used: Fine tuning of pre-trained LLM models(CamemBERT-bio, AliBERT, DrBERT) with LoRA, freezing the base model incur

1. Huge computational power
2. Expensive and slow
3. Need large datasets

Two light weight Transformer models are fine tuned using adapters outperform adapter augmented LLMs that are trained from scratch. This proved more efficient for the low in-resource, limited data for training and compute.

Datasets: Clinical notes dataset of France

Results: Accuracy is more when using GRN adapter for the Transformer models.

Performance comparison of transformer with different adapters.

Transformer	Acc (↑)	Pre (↑)	Rec (↑)	F1 (↑)	Training Time (hours) (↓)	Inference Time (s) (↓)
Baseline	0.85	0.85	0.83	0.84	0.11	3
Adapter	0.85	0.83	0.85	0.84	0.4	2
Lightweight	0.85	0.82	0.88	0.85	1	3
GRN	**0.87**	0.85	**0.89**	**0.87**	0.7	3
TinyAttention	0.85	0.81	0.88	0.84	0.7	3

Bold denotes the best values.

TABLE 5. Performance comparison of AliBERT with different adapters.

AliBERT	Acc (↑)	Pre (↑)	Rec (↑)	F1 (↑)	Training Time (hours) (↓)	Inference Time (s) (↓)
				Setup 1		
Baseline	0.86	**0.87**	0.84	0.86	39.3	128
Adapter	0.78	0.72	**0.88**	0.79	42.8	128
Lightweight	0.84	0.82	0.85	0.84	34.8	131
GRN	**0.87**	0.84	0.84	**0.85**	30.8	127
TinyAttention	0.84	0.81	0.83	0.82	42.2	128
				Setup 2		
Adapter	0.68	0.67	0.67	0.67	46.6	126
Lightweight	0.66	0.72	0.49	0.58	46.7	126
GRN	0.67	0.7	0.59	0.64	37.5	126
TinyAttention	0.67	0.67	**0.7**	0.68	46.2	130

Bold denotes the best values.

TABLE 6. Performance comparison of DrBERT with different adapters.

DrBERT	Acc (↑)	Pre (↑)	Rec (↑)	F1 (↑)	Training Time (hours) (↓)	Inference Time (s) (↓)
				Setup 1		
Baseline	**0.87**	0.84	**0.9**	**0.87**	45.2	133
Adapter	0.86	**0.87**	0.87	0.87	38.5	126
Lightweight	0.71	0.78	0.56	0.65	41.6	132
GRN	0.86	0.84	0.88	0.86	41.3	130
TinyAttention	0.85	0.81	0.9	0.85	28.9	123
				Setup 2		
Adapter	0.69	0.71	0.63	0.67	46.1	122
Lightweight	0.73	0.72	0.73	0.72	45	125
GRN	0.73	0.7	0.76	0.73	47	126
TinyAttention	0.75	0.76	0.69	0.72	47.5	123

Bold denotes the best values.

Conclusions: Transformer-based models trained from scratch perform comparably or better, especially in environments with limited computational resources and data availability. Among the adapter structures evaluated, the GRN demonstrated superior accuracy, precision, recall, and F1 score, making it the most effective adapter for enhancing clinical note classification.

Open Challenges:

1. With the use singleGPU,welackedthemem ory capacity and compute throughput required to fine-tune large proprietary models like Deepseek and Grok
2. Compared fully trainable versus fully frozen backbones with adapters; intermediate freezing ratios (e.g., 40-80% of layers frozen) may affect convergence speed and generalization, but were not explored.
3. Performance analysis focused primarily on accuracy and training time; other metrics such as memory footprint, latency, and energy consumption in diverse hospital server environments were not measured.

Relevance to Our Team: Hardware resource and the memory matters for precise computation during Query Answering in medical data.

Reference:

https://ieeexplore.ieee.org/abstract/document/11048527

Extracting Healthcare Quality Information from Unstructured Data

Problem Statement: Information extraction is a key Natural Language Processing (NLP) task for discovering and mining critical knowledge buried in unstructured clinical data. This paper talks about the software Canary, a free and open source solution designed for users without NLP and technical expertise and apply it to four tasks, aiming to measure the frequency of: (1) insulin decline; (2) statin medication decline; (3) adverse reactions to statins; and (3) bariatric surgery counselling. Canary has been developed for processing clinical documents to support the extraction of data using user-defined information discovery parameters and vocabulary.

Algorithms Used: The qualitative and quantitative research is applied in the Information Extraction of unstructured data.

- Define lexicon/vocabulary organised into different word classes
- Define grammatical rules to form the phrases out of word classes
- NLP's simple rules

Datasets: Clinical records of adult patients with Massachusetts General Hospital and Brigham and Women's Hospital between 2000 and 2014 for Insulin Decline and Statin decline case study.

Results: Insulin decline patients are evaluated using note level and sentence level.

	Sensitivity	Specificity	PPV
Note-level	100.0% (76.8–100.0)	99.9% (99.6–100)	93.3% (68.0–99.8)
Sentence-level	100.0% (82.4–100.0)	N/A	95.0% (74.4–99.9)

Conclusions: The principal advantage of our tool is that it is a GUI-based software which does not require any technical background. In this study, this was underlined by the fact that it was used by several researchers without any such technical background to successfully create language models of important clinical phenomena.

Open Challenges: The paper gives the quantitative study for different case study. Language models need to build for different clinical phenomena.

Relevance to Our Team: TF-IDF is used to calculate the Recall, Sensitivity and precision.

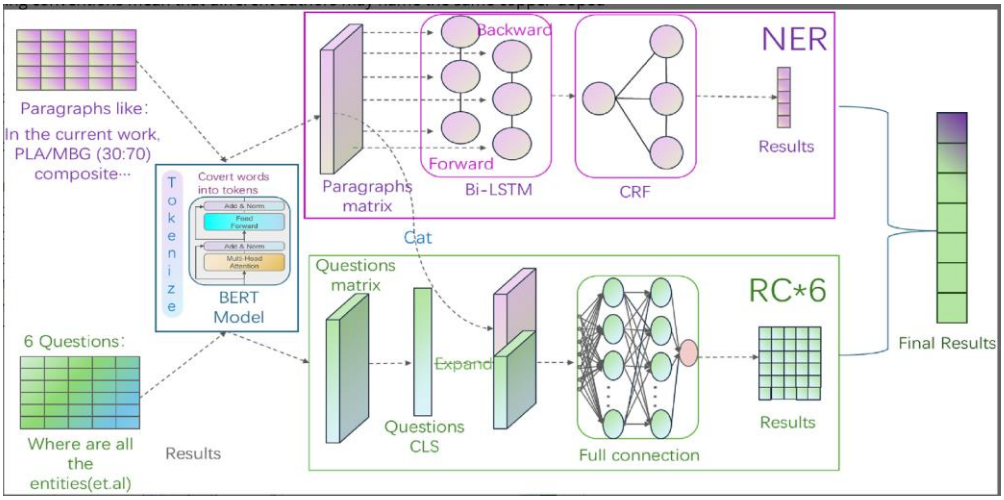
Reference:

https://pmc.ncbi.nlm.nih.gov/articles/PMC5977624/

<u>**Enhancing named entity recognition with a novel BERT-BiLSTM-CRF-RC joint training model for biomedical materials database**</u>

Problem Statement: This paper talks about a novel joint training model for named entity recognition (NER) that combines BERT, BiLSTM, CRF, and a reading comprehension (RC) mechanism. Traditional BERT-BiLSTM-CRF models often struggle with inaccurate boundary detection and excessive fragmentation of named entities due to their lack of specialized vocabulary. RC mechanism helps refine fragmented results by enabling the model to more precisely identify entity boundaries without relying on an expert-annotated dictionary.

Algorithms Used: BERT-BiLSTM-CRF-RC(reading comprehension) with CLS token for classifying article paragraph. Recognising entity boundary is a key to restore the semantic relationship in biomedical text.



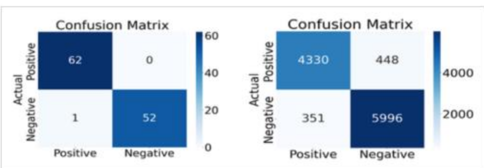
Datasets: Data from ScienceDirect website using the key mesoporous bioactive glass

Results:

TABLE 2. Parameters of models.

Parameter	Value
BERT hidden dimension	768
Learning rate for NER-RC and classifier	2e-5
Epsilon for Adam optimizer	1e-8
Dropout rate	0.1

Classifier	Acc	Precision	Recall	F1
	0.991	1.00	0.981	0.990



Conclusions: NER and RC systems can complement each other in practical applications, but they may also cause boundary misrecognition or redundancy.

As the amount of training data increases, the model's performance steadily improves, especially in terms of accuracy, recall, and F1 score, gradually approaching optimal values.

Open Challenges: Even with the use of reading comprehension to reinforce boundaries, the model still makes boundary errors during NER prediction, leading to subsequent cascading errors.

Relevance to Our Team: NER RC mechanism can be helpful to find the entity boundary in the medical text.

Reference:

https://onlinelibrary.wiley.com/doi/10.1002/mgea.70001

Enhanced Named Entity Recognition in Medical Texts Using Transformer-Based Models

Problem Statement: To extract the insights from the unstructured medical data, the approach leverages advanced Natural Language Processing techniques, specifically state-of-the-art transformer-based models like BioBERT and ClinicalBERT, to perform Named Entity Recognition in the medical domain. By integrating additional domain-specific resources, including comprehensive medical terminologies and ontologies, the performance of entity recognition is enhanced.

Algorithms Used:

1. Pre-processing: The raw text is pre-processed which involves Tokenization, Lowercasing, Removing Stop Words, Handling Abbreviations, Handling Special Characters.
2. Embeddings
    a. Domain specific embeddings using BioWord2Vec, ClinicalBERT
    b. Contextual embeddings using BioBERT, ClinicalBERT
3. BiLSTM-CRF for NER
4. Fine tuning using Spacy model
5. Output is the Annotated Text and Structured Data{Entity and Type}

Datasets: MTSamples dataset. The dataset consists of sample transcription of 40 different medical specialities. The dataset contains information of 940 medical problems, 227 treatements, and 209 tests.

Results: Metrics used to evaluate the model, where TP- true positive, FP- False Positive, TN- true negative, FN- False Negative, and I - total elements.

Conclusions: The transformer-based models like BERT out-performed the traditional models.

Open Challenges: The work can be looked for the multi-linguistic medical text.

Relevance to Our Team: BioBERT can be used for embeddings. Spacy model can be applied for NER which is helpful for question-answering

https://ieeexplore.ieee.org/document/10990753